Receiver-Grounded Link Abstraction for Trustworthy O-RAN Digital Twins: How Much Real PHY Does a Twin Need?

***Abstract*—Network digital twins are where O-RAN rApps are validated before they touch production, so the twin's link-to-system (L2S) abstraction—the map from SINR to throughput—decides which controllers look good. Received wisdom says this map must be measured on a real physical layer. We ask *how much* of a real PHY a twin needs, and answer by measuring two constants on OpenAirInterface (OAI). Our open, CPU-only, multi-cell twin (OpenStreetMap $\rightarrow$ Sionna RT $\rightarrow$ per-RE SINR $\rightarrow$ EESM $\rightarrow$ OAI's `nr_dlsim` curve) hosts a traffic-steering rApp and three bridges to a live OAI 5G Standalone stack. We separate a twin's *reporting* error from its *decision* error, the latter measured as rApp *regret*: the utility lost by planning on a surrogate twin and living in the OAI-grounded one. An uncalibrated Shannon curve inflates throughput 39.7% and costs 0.767 nats; a *fixed* 2 dB margin halves the first and barely touches the second; a single *fitted* margin δ=4.71 dB collapses both, to 1.6% and 0.033 nats—because OAI's staircase *is* a Shannon-optimal staircase displaced by δ, to 0.62 dB RMS. The customary "decisions differ" statistic (25–38$\%$ for every surrogate) separates none of them; only regret does. On the live stack the twin's link rate exceeds what OAI's MAC schedules by a *constant* factor (κ=0.470 $\pm$ 0.004, CV 0.8% over MCS 0–28), so it cancels in ratios: the twin's *relative* per-UE steering-gain prediction matches the real MAC to 1.2 ± 5.7 points. Delivered goodput lies a further 80% down and grows with SINR—the emulator's ceiling, not the twin's error. Finally, a null control on two *identical* real cells shows the natural goodput probe reporting $-21.0 \pm 1.4\%$ where truth is zero, retracting a multi-cell gap we previously reported.**

***Index Terms*—Digital twin, 5G NR, OpenAirInterface, link-to-system abstraction, O-RAN, rApp, traffic steering.**

I. Introduction

The digital twin is becoming the environment in which AI/ML-driven RAN control—O-RAN rApps and xApps—is trained and validated before it touches production [1]. Its value hinges on data fidelity: a traffic-steering rApp decides which cell serves each user from signal, interference and load, which depend jointly on 3D propagation, on neighbouring cells, and on a link model turning SINR into a real modulation-and-coding outcome. Evaluating such a controller needs all three, across many layouts rather than one scenario.

Open tooling splits into camps that do not meet: system-level simulators [2] scale to many cells but abstract the PHY, while OAI-based emulators run the real 5G-NR stack but, with realistic propagation, cover a single link (Table I).

A subtler gap concerns the *L2S abstraction* itself. Open simulators map SINR to a modulation-and-coding outcome through idealized or 3GPP-calibrated curves; we obtain ours from OAI's real receiver (`nr_dlsim`, LDPC decoding and rate matching). At 10 dB SINR it yields 2.16 bits/s/Hz against Shannon's 3.46—a 38% implementation loss—with non-uniform MCS steps spanning up to 2.2 dB.

The obvious claim is that this curve is therefore necessary. We tried to make it, and the evidence did not support it. The useful question is not *whether* an idealized curve is wrong—it plainly is—but *which part* of it a controller is sensitive to. That needs a metric this literature does not use: whether two twins pick different actions is not whether picking the wrong one costs anything. We measure *regret*. Under it, almost all of the OAI curve's value collapses onto one implementation margin δ; a second constant, a derate κ, converts the twin's link rate into what a real MAC schedules.

This paper contributes:

1) *Two constants, and evidence that they are enough.* We separate a twin's *reporting* error from its *decision* error (rApp regret). An uncalibrated Shannon curve inflates throughput 39.7% and costs 0.767 nats; a *fixed* 2 dB margin halves the first and barely moves the second; a *fitted* margin δ=4.71 dB collapses both (1.6%, 0.033 nats). It is δ's value, not a margin's presence, that matters, and the curve's shape buys nothing further. The customary "decisions differ" statistic is 25–38% for *every* surrogate; only regret distinguishes them.
2) *A constant, emulator-independent MAC derate.* On a live OAI Standalone stack at matched bandwidth, the twin's link rate overpredicts what OAI's MAC schedules by κ^{-1}, κ=0.470 $\pm$ 0.004—constant to 0.8% across MCS 0–28, of which the TDD duty 0.743 is exactly computable. Constancy is the point: κ cancels in ratios, so the twin's *relative* steering-gain prediction matches the real MAC to 1.2 $\pm$ 5.7 points, while delivered goodput sits a further 80% down and grows with SINR—the emulator's ceiling, not the twin's error. Conflating the two makes "fidelity gap" numbers irreproducible.
3) An open, CPU-only, multi-cell twin (Fig. 1) and three real-stack bridges, the third injecting *per-cell* ray-traced channels on a two-DU stack—the UE is the rfsimulator server, so DU-k's waveform is filtered by `rfsimu_channel_ue`k and the DUs' signals are physically summed: combined-waveform interference, not a noise proxy.
4) *A 24-run evaluation and two retractions.* A 3-city eval-

TABLE I
OPEN RAN DIGITAL-TWIN PLATFORMS ON FOUR AXES.

Platform	Real stack	Site RT	Multi cell	Ctrl loop
OWDT [3]	✓	✓	×	—
Tiny-Twin [4]	✓	$\times^a$	×	real
ns-O-RAN [5]	$\times^b$	$\times^b$	✓	model
OpenTwin [6]	$\times^b$	×	✓	model
OAI+FlexRIC [7]	✓	$\times^c$	✓	real
This work	✓	✓	✓	model*

[a]replayed traces; [b]ns-3 PHY, stochastic channel [2]; [c]over-the-air; *loop closed on the twin, executed as real F1 handovers on two live cells (Sec. VIII-D); the link dimension is grounded on a live SA stack (Sec. VIII-C).

uation with bootstrap CIs, per-scene and pooled paired tests, and an EESM-β control. We retract two claims that did not survive: a multi-cell fidelity gap—a null control on two *identical* cells reports $-21.0 \pm 1.4\%$ where truth is zero, and a controlled multi-cell measurement does not close on this emulator—and a correlation, as neither baseline outage nor peak load predicts rApp gain once computed *within* scene, where both reverse sign.

II. Related Work

Open RAN digital-twin platforms. Table I positions the closest open efforts on four axes; none combines all of them. Tiny-Twin [4] runs a real OAI stack with a RIC loop, single-cell from replayed traces. ns-O-RAN [5] and OpenTwin [6] close rApp loops across many cells but inherit ns-3's abstracted PHY and stochastic channel [2]. OAI+FlexRIC [7] closes handover loops over the air, without a ray-traced channel.

Positioning against OWDT. OWDT [3] is the closest work—OAI plus Sionna RT and a real PHY—and the distinction is architectural. It is *single-cell*, so with one transmitter its per-RE SINR reduces to an SNR and the association problem does not exist, whereas our engine carries a load-scaled interference sum over all non-serving cells (Eq. 2) and Bridge 3 makes that interference physically real. It is also *monitor-only*: nothing acts on the KPIs it streams. It can therefore report that a real PHY changes a KPI, but not what that change costs a decision—the question its design cannot pose.

Link-to-system abstraction. System-level simulation [8] relies on L2S mapping: a link-level simulator characterises SINR-to-BLER per MCS, and an effective-SINR mapping (EESM [9], [10]) collapses a frequency-selective SINR vector into an AWGN-equivalent value. Our lookup is measured from OAI's own PHY rather than from idealised curves. Our interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [11]: we derive each UE's interference from the ray-traced neighbour channels.

O-RAN control loops. The O-RAN architecture [12], [13] externalises RAN control: xApps on the near-RT RIC (via E2) and rApps on the non-RT RIC (via O1/A1). Traffic steering via cell-individual offsets (CIO) is a canonical rApp use case [1], which a digital twin lets one validate before deployment.

III. Digital Twin Architecture

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer covering every cell and UE, and a real-stack layer grounding sampled links in OAI. A WGS84 bounding box drives an OpenStreetMap query; footprints are extruded into a 3D Mitsuba scene with ITU materials. Cells and UEs come from a swappable source—a seeded random generator (N sites $\times$ 3 sectors, uniform UE drops) or a CSV reader for real cell-site coordinates. Sionna RT [14] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE, and computes the channel frequency response of every cell-to-UE link over an OFDM resource grid; propagation settings and their rationale are in the reproducibility statement.

IV. OAI-Grounded KPI Engine

Let $\mathbf{H}_{u,c}(k)$ be the ray-traced channel of link $c \to u$ on resource element (RE) k, P_c^{tx} the cell transmit power, and $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \overline{\|\mathbf{H}_{u,c}\|_F^2}$. Each UE associates to the cell maximising it, biased by a cell-individual offset (CIO)—the O-RAN control knob:

$$s(u) = \arg\max_c \; P_{u,c}^{\text{rx}} \cdot 10^{\,\text{CIO}_c/10}. \tag{1}$$

Under airtime sharing each cell serves one UE at a time, so the optimal single-stream precoder is the matched filter and the serving per-RE gain is $\sigma_{\max}$. With flat per-RE power p_c, neighbour load $\rho \in [0,1]$ and thermal noise $N_0 = kTB \cdot \text{NF}$:

$$\gamma_u(k) = \frac{p_s\, \sigma_{\max}\big(\mathbf{H}_{u,s}(k)\big)^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \tag{2}$$

EESM compresses the per-RE SINRs to one effective value, preserving the ray tracer's frequency selectivity:

$$\gamma_u^{\text{eff}} = -\beta \, \ln\Big(\tfrac{1}{K} \textstyle\sum_k e^{-\gamma_u(k)/\beta}\Big), \tag{3}$$

with β set per modulation order [10] and one scale parameter. These are literature defaults, not values fitted to `nr_dlsim` under frequency-selective channels; rather than assume that benign we bound it in Sec. VIII-B—a bound on magnitude, not on structural error across modulation orders.

OAI-measured link curve. The effective SINR maps to an MCS and spectral efficiency, $(m_u, \eta_u) = \Phi_{\text{OAI}}(\gamma_u^{\text{eff}})$, through a curve *measured offline from OAI's* `nr_dlsim`, so it carries a real receiver's losses.

At 10 dB it delivers 2.16 bits/s/Hz—38% below Shannon's 3.46—and the QPSK→16QAM boundary spans 2.2 dB against ∼1 dB for adjacent QPSK steps. Whether that *shape* matters to a controller, as opposed to the 38% level, is what Sec. VIII-B answers.

On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s}\, \eta_u \,(1 - \text{BLER}_{\text{tgt}}), \tag{4}$$

where B_s is the serving cell bandwidth and K_s its number of attached UEs.

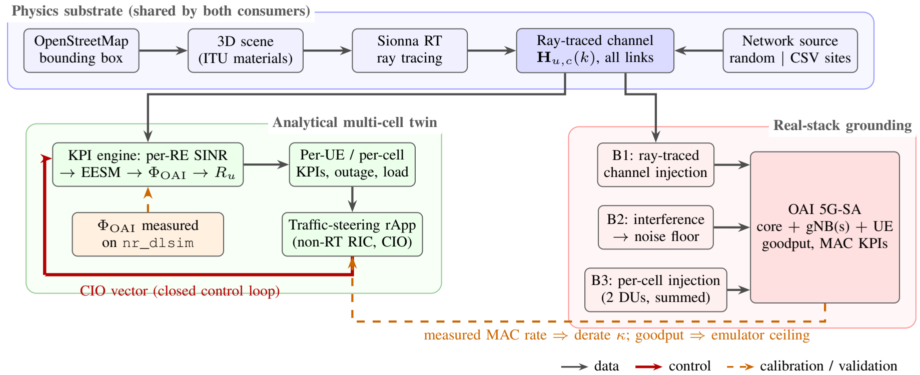

Fig. 1. System architecture. One physics substrate produces the ray-traced channel for every cell–UE link; two consumers share it. In the analytical twin, Φ_{OAI} grounds throughput and the rApp closes a CIO loop. In the real-stack layer, three bridges drive a live OAI 5G Standalone system: comparing the twin against what OAI's MAC *schedules* yields the constant derate κ, while comparing against delivered goodput additionally exposes the emulator's ceiling.

V. Traffic-Steering rApp

The rApp adjusts the CIO vector to maximise the proportional-fair utility $U(\mathrm{CIO}) = \sum_u \log R_u$. Because U is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus. We therefore do coordinate ascent: starting from $\mathrm{CIO} = \mathbf{0}$, sweep the cells, line-searching each cell's CIO over a grid while every *other* cell is held at its current best value (a true coordinate ascent, not a reset-to-zero sweep), and keep the maximiser of U; stop when no cell's line search improves U. This is monotonically non-decreasing in U by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

One layout converges in 2–3 sweeps (9 cells, CIO grid $\{-12..+12\}$ dB), 4.7 s of coordinate ascent. The rApp is a *reference* controller, not a contribution; every evaluation reuses the same engine, so the loop rides on the exact twin used for the baseline, and the output handover list is executable via O1/telnet at the non-RT RIC timescale.

VI. Twin-to-Real-Stack Bridges

Three bridges connect the analytical twin to a live OAI stack. Their status, stated up front: Bridge 1 is validated and carries every measured fidelity number here; Bridge 2 is a design specification, *not* a demonstrated result; Bridge 3's injection is verified on the stack, but the multi-cell measurement it enables does not close (Sec. VIII-D). All measured fidelity results reflect the single-gNB link dimension only.

Bridge 1: channel injection. A link's ray-traced CFR becomes a sample-spaced impulse response (IFFT); the strongest causal taps are kept, unit-energy normalised, mapped to sample indices, and override the rfsimulator's channel via a patch to OAI (`oai_rt_inject_channel`), confirmed in the gNB log.

Bridge 2: interference-as-noise-floor. A single-gNB link has no notion of the other cells a UE hears. Per UE, the twin's inter-cell interference becomes a noise-floor rise $\Delta N_u = 10\log_{10}((I_u + N_0)/N_0)$, written as a per-client OAI noise power—a site-specific generalisation of OCNG [11].

Bridge 3: per-cell injection. On a two-DU stack the UE is the rfsimulator *server*, so DU-k's samples pass through `rfsimu_channel_ue`k and `rxAddInput` sums both waveforms. Injecting each UE–cell link's taps there makes the non-serving cell physically combined interference, not Bridge 2's noise-floor proxy.

VII. Experimental Setup

Scenes and layouts. Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry built once per scene, reused across eight seeded layouts each (24 runs). 3.5 GHz, $B = 38.16$ MHz (106 PRB at 30 kHz SCS, matching the live OAI cell), 46 dBm per cell, 4×1 BS array, 7 dB noise figure, depth 3, 9 cells and 30 UEs per layout, $\rho = 1.0$. The CFR is exported over all 1272 subcarriers, so the per-RE SINR and its EESM compression cover the cell's full band; a sweep varies $\rho \in \{0, \dots, 1\}$ on a cached channel.

Metrics. Per-UE downlink throughput (median, mean); *served cell-edge* rate, the 5th percentile among UEs with non-zero throughput (a raw 5th percentile collapses to zero once outage exceeds 5%); *outage*, the fraction with zero throughput; Jain's index; and *peak cell load reduction*, the decrease in the maximum UEs attached to any single cell (negative = load shed). Gains carry seeded bootstrap 95% CIs and paired tests.

Clustering. The eight seeds of a scene share one geometry, so the pooled n=24 tests have a cluster structure and overstate the effective sample size. We therefore report the three per-scene tests (n=8 independent layouts each) alongside the pooled one, and treat per-scene directional consistency, not the pooled p alone, as the evidence.

VIII. Results

A. Traffic-steering rApp

Across 24 runs (Table II) the rApp helps the users a strongest-cell policy under-serves and sheds peak load: median per-UE rate rises $15.8 \to 18.5$ Mbps, served cell-edge $5.8 \to 7.4$ Mbps, both significant on the pooled absolute rates (paired Wilcoxon), as is the peak-load reduction. Because seeds within a scene share geometry, we also test per scene (n=8): median

TABLE II

RAPP GAINS VS. STRONGEST-CELL ASSOCIATION, 24 RUNS. MEAN WITH A SEEDED PERCENTILE-BOOTSTRAP 95% CI; p FROM A PAIRED WILCOXON SIGNED-RANK TEST ON THE ABSOLUTE RATES (SIGN TEST FOR JAIN); "+/−/T" = POSITIVE/NEGATIVE/TIED RUNS.

Metric	Overall (95% CI)	p	+/−/t
Median tput	$+10.9$ $[6.8, 14.8]$	$2{\cdot}10^{-4}$	19/3/2
Served edge	$+25.9$ $[15.8, 36.9]$	$4{\cdot}10^{-4}$	16/3/5
Jain index	$+6.2$ $[1.0, 11.6]$	0.052	16/6/2
Peak load (UE)	-0.62 $[-0.96, -0.29]$	$6{\cdot}10^{-3}$	—
Mean tput	$+3.7$ $[1.2, 6.4]$	0.036	15/7/2
Outage	0.0 (all runs)	n/a	0/0/24

rate reaches $p<0.05$ in Manhattan and Paris but not Berlin (0.148); served cell-edge in Berlin and Manhattan but not Paris (0.156). The improvement is directionally consistent across all three geometries; the pooled p corroborates, it is not 24 independent samples. Three caveats. Aggregate rate rises only 3.7%; Jain's index (+6.2%) misses significance (p=0.052) and is *negative* in Paris. Outage is bit-for-bit identical in all 24 runs, so no test applies: steering relocates covered users, it does not create coverage. On 2–5 layouts the rApp moves no UE, an exactly zero gain; those ties are excluded from the sign test (Dixon–Mood).

Per scene: Berlin, the densest, has the lowest outage (10.4%) and sheds the most peak load (-1.0 UEs) but gains least in median rate; Paris has the highest outage (25.8%) and loses fairness; Manhattan gains most in fairness (+17.5%). The direction is robust across all three, the magnitude strongly deployment-dependent—what a single-scenario study would misrepresent.

B. What a twin says vs. what acting on it costs

The 38% link-curve gap of Sec. IV is real, but does a controller care? Hold geometry, ray tracing, association and interference fixed; change *only* the link curve. The *reporting* error is how far a surrogate twin's predicted throughput sits from the OAI-grounded twin's. The *decision* error is *regret*: evaluate the surrogate's chosen CIO on the OAI-grounded twin and compare with the CIO it would have chosen had it planned there: $\mathcal{R} = U(\mathrm{CIO}^{\star}_{\Phi_{\mathrm{OAI}}} \mid \Phi_{\mathrm{OAI}}) - U(\mathrm{CIO}^{\star}_{\Phi_{\mathrm{sur}}} \mid \Phi_{\mathrm{OAI}}) \geq 0$, with U the proportional-fair objective the rApp maximises. Reporting error is what you get wrong reading a number off the twin; regret is what you pay for acting on it. We compare four surrogates over the same MCS/SE set, differing only in the SINR at which each MCS becomes available: the Shannon-optimal staircase; a *fixed* 2 dB margin, as a link-budget abstraction assumes; a *fitted* offset $\Phi(\gamma) = \log_2(1 + \gamma/\delta)$, $\delta = 4.71$ dB; and a fitted attenuation $a\log_2(1+\gamma)$, $a = 0.606$. The last three are the honest baselines—the Shannon staircase is a straw man no simulator uses.

The two errors are close to independent (Fig. 2). The idealized curve inflates throughput 39.7% $[30.4, 49.9]$ and costs 0.767 nats. A *fixed* 2 dB margin halves the reporting error (+24.4%) but barely touches the regret (0.732). The *fitted* margin δ removes both—+1.6% $[-0.1, 3.5]$ and 0.033 $[0.006, 0.071]$ nats, a 23× reduction—so it is the *value* of δ, not a margin's presence, that matters. The attenuation model reports well (-5.7%) yet is as costly to act on as no calibration at all (0.800 nats): reporting accuracy does not imply decision quality. Most tellingly, the statistic this literature usually reports—the fraction of layouts on which the two twins select different steering actions—is 37.5%, 29.2%, 25.0% and 37.5% across the four surrogates. It cannot tell the good one from the bad. Nor can the Jaccard overlap it is usually paired with, which has no natural null. Regret has one: zero.

Why one scalar suffices is visible in Fig. 2 (left): OAI's staircase *is* a Shannon-optimal staircase displaced by δ. Its MCS thresholds sit 0.62 dB RMS (1.14 dB max) from the fitted-offset curve across MCS 0–28, against 4.75 dB uncalibrated and 2.78 dB for the assumed 2 dB margin. The attenuation model is the instructive failure: 2.63 dB RMS but 6.19 dB max, its error concentrated at the extremes where cell-edge and cell-centre users live—which is why it reports well and acts badly. What the twin needs from a real PHY is therefore δ; the curve's non-uniform MCS spacing does not change what the controller does.

As a control on the twin's own free parameter, perturbing the EESM β scale by ±20% shifts the mean effective SINR by under 0.15 dB, changes decisions on 4–8% of layouts, and costs at most 0.046 nats—the same order as the residual regret of the best-calibrated curve, ∼17× below an uncalibrated one. The conclusions are not an artefact of that knob.

C. The real fidelity gap

The analysis above compares two *models*; we now close the loop on the real stack. A full OAI 5G Standalone system runs on the same CPU host. On a single gNB we ground the *link* dimension: pin the downlink operating point by capping the scheduler MCS over a clean channel, sweep the cap across MCS 0–28, and measure goodput at each fixed MCS, which indexes the same OAI curve the twin uses. Two details decide the answer. The live cell is 106 PRB at 30 kHz SCS, i.e. $B = 38.16$ MHz occupied, not the twin scenario's 20 MHz, so the twin's rate must be evaluated at the cell's own bandwidth. And a saturating UDP probe is unusable: in reverse mode `iperf3` reports the *sending* rate, so with the scheduler pinned to MCS 4 (a 9.8 Mbps MAC) an 8 Mbps offer reads exactly 8.00 Mbps at ∼0% loss. TCP cannot exceed the channel but needs tens of seconds to leave slow start, so we settle and measure its plateau.

Two references matter, and conflating them makes "fidelity gap" numbers irreproducible. Against **what OAI's MAC schedules**—real scheduler code paying TDD duty, DMRS, PDCCH, HARQ—the twin's link rate $B\,\eta_{\mathrm{OAI}}(1{-}\mathrm{BLER})$ is high by a *constant* factor: $\kappa = 0.470 \pm 0.004$ (52.9% inflation, ranging only 52.4–53.6% over MCS 0–28; CV 0.8%). Against **delivered goodput** it is high by a median 80%, and that gap is *not* constant: 72% at MCS 0 to 87% at MCS 28 (CV 27%). At MCS 28 the twin predicts 190.8 Mbps, TDD duty allows 141.7, the MAC schedules 88.9, the application

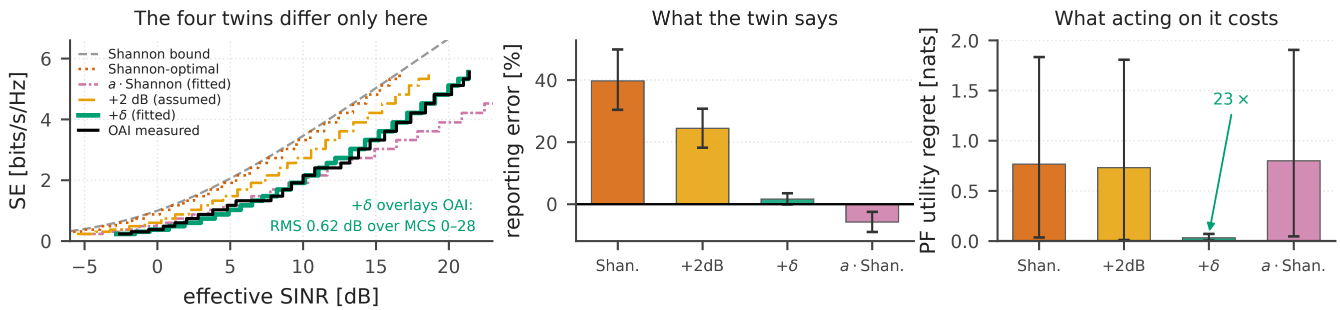

Fig. 2. **Left:** the four twins share one $(\mathrm{MCS}, \mathrm{SE})$ set and differ *only* in the SINR at which each MCS unlocks; the fitted-offset curve overlays the OAI staircase (0.62 dB RMS), while the fitted attenuation tracks it on average and diverges at both ends. **Centre, right:** reporting error and regret, means with seeded bootstrap 95% CIs over 24 runs. The two are close to independent—the attenuation model reports best of the three surrogates and acts worst.

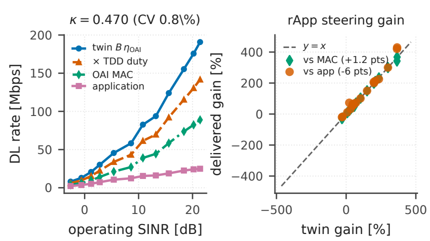

Fig. 3. Single gNB, matched bandwidth. Left: at matched operating SINR—the twin's link-level rate, that rate scaled by the TDD DL duty, OAI's scheduled MAC throughput, and delivered goodput. The twin-to-MAC ratio is constant ($\kappa = 0.470 \pm 0.004$); the MAC-to-application spread is the emulator's ceiling and grows with SINR. Right: per-UE rApp steering gain. Against the MAC the twin sits on $y{=}x$ ($+1.2 \pm 5.7$ pts); against delivered goodput it does not.

receives 24.9. Only the first two steps are modelling: the TDD duty ($104/140 = 0.743$) and the MAC's physical-channel and scheduler overhead (0.633) account for the *whole* twin-to-MAC discrepancy—which is why κ is constant. Everything below the MAC line is an *upper bound* on unmodelled stack effects: it mixes the rfsimulator's compute ceiling, same-host contention and transport, which one machine cannot separate. We quote κ, not the 80% (Fig. 3, left).

Constancy is not a detail: it *is* the second constant. A multiplicative κ cancels in any ratio, so it should not perturb the twin's *relative* predictions. We test that by propagating each curve through the rApp over eight layouts (240 per-UE evaluations, 19 steered UEs), indexing every curve by the twin's own operating MCS—the twin's rate is a staircase in SINR, so interpolating a measured curve along SINR would compare a staircase with a ramp and manufacture a gap between the knots. The airtime share is kept in *both* rates, since it is most of the rApp's benefit and cancels in the difference. Against the MAC the twin is right: 126.6% predicted vs 125.4% scheduled, a gap of $+1.2 \pm 5.7$ points whose bootstrap 95% CI $[-0.7, +4.1]$ straddles zero (Fig. 3, right). Against delivered goodput it is not: median -6 points, CI $[-22.2, -4.3]$, excluding zero—that residual is the emulator ceiling, not the abstraction.

D. Multi-cell closed loop, and what it does not yet measure

Two real cells (CU plus two F1 DUs) match a two-sector twin; the rApp's decision executes as a real inter-cell handover via scripted CU telnet, PDU session preserved: the loop closes. *Measuring* the delivered gain is another matter. Our first attempt reported the twin's per-UE gain (66%) overshooting the delivered gain (9.0%) by 57 ± 165 points, and blamed the DUs' clean channel. That does not survive a null control: give both cells the *same* channel and the true effect is exactly zero, so whatever the harness reports is bias. It reports $-21.0 \pm 1.4\%$. The cause is timing—a 6 s TCP flow started 3 s after each handover, inside both slow start and the link-adaptation ramp—and since `trigger_f1_ho` always moves DU0→DU1, measurement order and cell identity were confounded, so the reported sign followed whichever direction the rApp chose (mean DL MCS 16.5 on PCI 0 vs 10.7 on PCI 1). Probing at TCP's plateau gives $-2.4{\pm}2.9\%$, consistent with zero, MCS 28 on both. We withdraw the $57{\pm}165$ number.

With the instrument fixed we ground the channel (Bridge 3). Injection works: every layout's UE log confirms both per-cell models carrying the twin's ray-traced gains, and the UE reaches MCS 27 on the stronger cell. Two obstacles remain, neither the twin's. With channel models on both rfsim connections the loop destabilises—over three layouts two F1 handovers never complete, the third only through 70 UE re-synchronisations. And rfsim's noise floor is a config knob, not $kTB \cdot \mathrm{NF}$: injection sets the *relative* per-cell gain, and sweeping the floor -22 to -2 dB gives operating MCS $8, 19, 14, 18, 14, 0$—non-monotone, hence not invertible. A controlled multi-cell gap needs absolute operating-point matching, which this emulator cannot deliver.

E. Coverage, outage, and load sensitivity

Mean UE outage is 17.8% at full neighbour load. What predicts a layout's rApp gain? Pooled over 24 runs, median gain correlates weakly with peak cell load ($r{=}0.29$) and negligibly with baseline outage ($r{=}0.05$)—but a correlation over three cities can be produced by the between-city contrast alone. Recomputed *within* scene, neither survives: peak load gives $r{=}{+}0.46$, -0.08, $+0.18$ and outage -0.42, -0.29,

$+0.08$ (Berlin, Paris, Manhattan), changing sign. **We claim neither predictor.** An earlier version—sub-band CFR, band-wide EESM, unseeded solver—reported a confident $r{=}-0.42$ on outage; the full band and a seeded solver dissolved it.

What does hold is the regime separation: lowering ρ from 1.0 to 0 raises mean throughput 1.68–$1.95\times$, separating a coverage-limited tail steering cannot fix from an interference-limited body where load balancing acts.

IX. Discussion and Limitations

We state these plainly. (1) The loop is closed on one gNB and on two cells with scripted F1 handovers; autonomous E2-RC steering is future work, and Bridge 2 awaits multi-cell rfsim validation, carrying no fidelity claim here. (2) The single-gNB numbers are measured against *this* emulator, which delivers only 24.9 of the 88.9 Mbps its own MAC schedules at MCS 28; the twin's error is bounded by the MAC line, which is why we headline κ. (3) Bridge 3's injection is verified and rfsim sums the DUs' waveforms, superseding Bridge 2's noise-floor rise—but the multi-cell measurement does not close (Sec. VIII-D), and the injected taps carry each link's gain, not its delay profile. (4) The two constants are established for *one* controller on a proportional-fair objective, on a static full-buffer downlink snapshot; whether one scalar δ suffices for controllers reading MCS directly is open, and mobility, uplink and field calibration are next.

Guidelines. (i) *Calibrate the SINR axis; do not assume a margin*: one link-level sweep buys δ and cuts regret $23\times$, a round 2 dB margin buys almost nothing. (ii) *Measure regret, not decision divergence*, which was 25–38% for every surrogate. (iii) *Compare against what the MAC schedules, not application goodput*: below the MAC lies the emulator's ceiling. (iv) *Run a null control before reporting any twin-versus-real gap*: ours read $-21.0 \pm 1.4\%$ where truth is zero. (v) *Report correlations within scene*; pooled over three cities, ours reversed sign.

X. Conclusion

We asked how much of a real physical layer a 5G digital twin needs. For a proportional-fair traffic-steering rApp, two constants suffice. A *fitted* margin $\delta{=}4.71$ dB cuts an uncalibrated twin's reporting error from 39.7% to 1.6% and its regret from 0.767 to 0.033 nats—while a merely *assumed* 2 dB margin does not (0.732), so it is δ's value, not a margin's presence, that matters. A derate $\kappa{=}0.470 \pm 0.004$, constant to 0.8% across MCS 0–28, converts the twin's link rate into what a real 5G MAC schedules; being constant it cancels in ratios, so the twin's *relative* steering-gain predictions are already right to 1.2 ± 5.7 points. Below the MAC lies an 80% shortfall growing with SINR: the software radio's ceiling, not the twin's error. Two methods earn their place beside the constants: "decisions differ" cannot separate a faithful twin from an unfaithful one, only regret can; and a twin-versus-real comparison must be run against a null control—how we came to retract a multi-cell gap we had published.

Reproducibility

Code and data: Apache-2.0. **Software:** OAI `develop@0dab3bb` (tag `2026.w27`); Sionna RT 2.0.1. Φ_{OAI}: sweep SNR $\times$ MCS through `nr_dlsim` (106 RB, 12 frames/point), taking at 0.2 dB granularity the lowest SNR whose first-transmission BLER meets the 10% target (-2.6 to 21.4 dB). Bridge 1 keeps the 4 strongest causal taps, unit-energy normalised, gain in `path_loss_dB`; Bridge 3 keeps one tap (38.16 MHz resolves delay only to 26 ns). **Propagation:** specular reflection and refraction, depth 3; diffuse scattering and edge diffraction disabled, so we claim no diffuse-scattering fidelity. The solver is seeded from the layout seed: its path sampling is stochastic and an unseeded solve drifts every downstream number by $\sim 1\%$. **Runtime:** a layout costs 13.1 s ray tracing $+\,4.7$ s coordinate ascent; the 24-run sweep plus load sweep, 703 s, CPU-only. **Rebuilding:** `paper_stats.py` recomputes every statistic here from committed CSV/JSON and `paper_figures.py` every figure; `build.sh` fails if page count, overfull boxes or undefined references regress.

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